6 **(a)** **(i)** Explain what is meant by the term *nuclear binding energy*.

- The binding energy of a nucleus is the minimum energy required to completely separate to infinity all the nucleons of a nucleus. [B1]
- The binding energy is the energy equivalent of the mass defect Δm of the nucleus: binding energy = $(\Delta m)\, c^2$ [B1]

.....[2]

(ii) A nucleus of strontium-90 has a mass of 89.907 *u*. It contains 38 protons. The mass of a neutron is 1.00867 *u* and the mass of a proton is 1.00783 *u*.

Calculate the nuclear binding energy per nucleon of strontium-90.

mass of 38 protons = 38.2974; mass of 52 neutrons = 52.4508; total 90.7484	
mass defect = 90.7484 – 89.907 = 0.841 *u*	[B1]
$0.841\ u \times 1.66 \times 10^{-27}\ \text{kg} \times (3.00 \times 10^{8})^2$ $= 1.26 \times 10^{-10}$ J $= 1.40 \times 10^{-12}$ J per nucleon (3 s.f.)	[B1]

binding energy per nucleon = ... J [2]

(b) The nuclear fission of uranium-235 can be described by

$$^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3{}^{1}_{0}n.$$

Table 6.1 shows the masses of the particles involved in this reaction.

Table 6.1

particle	mass/u
$^{1}_{0}n$	1.008665
$^{89}_{36}Kr$	88.917636
$^{144}_{56}Ba$	143.922953
$^{235}_{92}U$	235.043930

Determine the energy released by the fission of 15.0 kg of uranium-235.

mass defect, m = [235.043930 – (143.922953 + 88.917636 + (2×1.008665))] u = 0.186011 u	[B1]
$E = m c^2$ and m = (mass in u) × (1.66×10^{-27} kg u^{-1}) = [$0.186011 \times 1.66 \times 10^{-27}$] × ($3.00 \times 10^{8}$ m s^{-1})^2 = 2.78×10^{-11} J	[B1]
Number of atoms, $N = N_A \times \frac{\text{mass of sample}}{\text{molar mass}}$ or $\frac{\text{mass of sample}}{\text{mass of 235 nucleons}}$ $= 6.02 \times 10^{23}\ \text{mol}^{-1} \times \frac{15000\ \text{g}}{235\ \text{g mol}^{-1}}$ or $\frac{15.0\ \text{kg}}{235 \times 1.66 \times 10^{-27}\ \text{kg}}$ $= 3.84 \times 10^{25}$ or 3.85×10^{25}	[B1]
energy released = $(3.85 \times 10^{25})(2.78 \times 10^{-11}\ \text{J}) = 1.1 \times 10^{15}\ \text{J}$	[B1]

energy = ... J [4]

[Total: 8]